

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Basic Trigonometry

Choose the triangle rule from the sides and angle you can see.

BASIC TRIGONOMETRY · P1 · 09

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA11 website answer bank. The topic reduces to these exam moves.

01 RATIO: Right-Triangle Ratios

TRIGGER *right-angled triangle*

02 SINE: Sine Rule

TRIGGER *non-right triangle with matching side-angle pair*

03 COS: Cosine Rule

TRIGGER *two sides and included angle, or three sides*

04 AREA: Triangle Area

TRIGGER *two sides and included angle*

BANK EVIDENCE Local pages: 29 problem, 55 answer, 14 notes. Website primary entries: 6.

01 RATIO: Right-Triangle Ratios

TRIGGER *right-angled triangle*

BECOMES Label opposite, adjacent, hypotenuse.

FIRST LINE TO WRITE

$$\sin \theta = \frac{O}{H}, \quad \cos \theta = \frac{A}{H}, \quad \tan \theta = \frac{O}{A}$$

SIMPLEST STRATEGY

- 1 Read the angle.
- 2 Assign O/A/H.
- 3 Take the ratio with the knowns.
- 4 Insert values.
- 5 Obtain side or angle.

WORKED MODEL

$$\sin 30^\circ = \frac{x}{10} \Rightarrow x = 5.$$

HARD VARIANTS

- 1 In compound diagrams, split the shape into right triangles before using SOH CAH TOA.
- 2 If the angle is not the reference angle, relabel opposite and adjacent from the correct angle.
- 3 Use exact surd values for special angles when the question asks for exact form.

— BOTTOM LINE

SOH CAH TOA starts with labels, not the calculator.

02 SINE: Sine Rule

TRIGGER *non-right triangle with matching side-angle pair*

BECOMES Use side over sine of opposite angle.

FIRST LINE TO WRITE

$$\frac{a}{\sin A} = \frac{b}{\sin B}$$

SIMPLEST STRATEGY

- 1 Spot an opposite pair.
- 2 Insert known side-angle pair.
- 3 Name the unknown pair.
- 4 Evaluate carefully.

WORKED MODEL

$$\frac{x}{\sin 40^\circ} = \frac{8}{\sin 70^\circ}.$$

HARD VARIANTS

- 1 Ambiguous SSA cases can give two possible angles; check the diagram and angle sum.
- 2 Pair each side only with its opposite angle.
- 3 After finding an angle, confirm the triangle angle sum is possible.

— BOTTOM LINE

Sine rule needs opposite pairs.

03 COS: Cosine Rule

TRIGGER *two sides and included angle, or three sides*

BECOMES Use the formula with the opposite side first.

FIRST LINE TO WRITE

$$a^2 = b^2 + c^2 - 2bc \cos A$$

SIMPLEST STRATEGY

- 1 Choose the side opposite the angle.
- 2 Organise b, c, A .
- 3 Substitute into cosine rule.

WORKED MODEL

$$a^2 = 5^2 + 7^2 - 2(5)(7) \cos 60^\circ.$$

HARD VARIANTS

- 1 Use cosine rule for SAS or SSS cases, especially when Pythagoras almost fits but no right angle is given.
- 2 When finding an angle, rearrange for $\cos A$ before applying inverse cosine.
- 3 Check whether the answer should be acute or obtuse from the side lengths.

— BOTTOM LINE

Cosine rule replaces Pythagoras when the angle is not 90° .

04 AREA: Triangle Area

TRIGGER *two sides and included angle*

BECOMES Use $\frac{1}{2}ab \sin C$.

FIRST LINE TO WRITE

$$\text{Area} = \frac{1}{2}ab \sin C$$

SIMPLEST STRATEGY

- 1 Assign the two sides around the angle.
- 2 Read the included angle.
- 3 Evaluate $\frac{1}{2}ab \sin C$.
- 4 Add units squared.

WORKED MODEL

$$\text{Area} = \frac{1}{2}(6)(9) \sin 40^\circ.$$

HARD VARIANTS

- 1 The angle must be included between the two sides used in $\frac{1}{2}ab \sin C$.
- 2 For compound areas, subtract or add triangle areas after finding missing sides.
- 3 Give square units and keep exact form if requested.

— BOTTOM LINE

The angle must sit between the two sides.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
right-angled triangle	$\sin \theta = \frac{O}{H}, \quad \cos \theta = \frac{A}{H}, \quad \tan \theta = \frac{O}{A}$
non-right triangle with matching side-angle pair	$\frac{a}{\sin A} = \frac{b}{\sin B}$
two sides and included angle, or three sides	$a^2 = b^2 + c^2 - 2bc \cos A$
two sides and included angle	$\text{Area} = \frac{1}{2}ab \sin C$